

Modeling and integration of water desalination units in thermal unit commitment considering energy and water storage

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ARTICLE INFO

Keywords:

Battery energy storage, mixed integer linear programming, short-term scheduling
Thermal power plant
Water desalination
Water storage

ABSTRACT

In this paper, a new short-term daily scheduling model is proposed and formulated. In the proposed model, thermal power plant will be optimally scheduled in a system with water desalination units. For the thermal units, various cost terms including start-up, shut-down, and also fuel cost are considered and modeled. Also, thermal dynamics of the plants including power generation ramp-up and ramp-down, minimum up and down time limits, and start-up and shut-down ramps are considered. In addition, energy storage batteries as well as water storage tanks of the desalination units are modeled. The proposed model is a mixed integer linear programming (MILP) for simultaneous scheduling of thermal power plants and desalination units while considering transmission constraints. Power and water balance at the network buses in addition to the power flow constraints of the lines are considered. Various membrane and thermal desalination methods are modeled. The proposed model is implemented on a test case and the simulation results show a significant reduction in the system operating costs as well as improved performance of both the thermal power plants and the desalination unit. The proposed scheduling method can be used for the various future studies of joint power-water production and coordination.

1. Introduction

More than 95% of the water on Earth is not potable [1]. The remaining 5% of drinking water is also not fully available for consumption [2]. The crisis of drinking water supply is more acute in arid and semi-arid countries. Other countries in the world have also come up with different ways of producing, reusing or saving water in the past decade [3]. Fig. 1 shows the estimated sources of drinking water in the world in 2050 [4]. As it can be seen, water desalination and water consumption reduction in agriculture are equal to each other and with a great difference from the other alternatives. Construction of new reservoirs and recycling urban and industrial waste waters are also the next sources.

Water supply reduction through agricultural consumption management is subject to various factors such as local conditions, weather conditions, types of crops and cannot be considered as a major drinking water source in the long run [5]. But in the past decade, water desalination has been considered as a major alternative to potable water production, especially in arid countries. Water desalination refers to all processes that reduce the amount of salt in the saline water. These methods can generally be classified into two types of membrane and thermal methods [6]. Reverse Osmosis (RO) is one of the most important membrane methods as well as the most widespread desalination

method [7]. In contrast, the most well-known thermal methods are the Multi-Stage Flash (MSF) and the Multi-Effect Desalination (MED).

One of the problems with water desalination in different ways is the high energy consumption. Although many efforts have been made so far to reduce the energy consumption of these units by introducing new methods of desalination [8] or modifying previous processes [9], in practice the problem of energy consumption has not been fully resolved [10]. Given that a very high percentage of the energy required by these units comes from fossil fuel units, high energy consumption will also increase greenhouse gas emissions [11]. In recent years, the use of renewable energy sources as one of the strategies to reduce energy consumption and air pollution has been simultaneously studied [12]. But it should be noted that at present, most of the energy required for these units is still provided by connecting to the grid and through fossil fuel thermal units. The generation planning of these thermal units is mainly short term and is called unit commitment (UC) [13]. Depending on the thermal dynamics of these units, turning the units on or off as well as changing the amount of power generation (increase or decrease) is subject to the certain functions and constraints. Accordingly, the output of the unit commitment problem is to determine the on/off status as well as the generation rate of these units optimally. The UC model should consider thermal dynamics specific to each generation unit. The final goal of the scheduling is minimizing the total cost of operation

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<https://doi.org/10.1016/j.desal.2020.114411>

Received 22 December 2019; Received in revised form 21 February 2020; Accepted 27 February 2020

Available online 09 March 2020

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Nomenclature**Sets**

Ω_H	time periods
Ω_B	battery energy storage systems
Ω_D	water desalination units
Ω_N	thermal power plants
Ω_I	network buses
Ω_M	auxiliary variables used for linearization

Parameters

$BL_{(i,j)}$	susceptance of transmission lines between buses i and j (PU)
$C_{(n)}^{On}$	unit start-up cost of power plant n (\$)
$C_{(n)}^{Off}$	unit shut-down cost of power plant n (\$)
$E_{(s,h)}^{BSS}$	stored energy in the battery storage s at time period h (MWh)
$EB_{(s)}^{Cha}$	charging efficiency of the battery storage s
$EB_{(s)}^{Dis}$	discharging efficiency of the battery storage s
$ER_{(s)}^{BSS}$	energy capacity of the battery storage s (MWh)
K_{Joule}^{kWh}	conversion factor from mega-joule to kilowatt-hour
k_n	constant cost coefficient of the power generation cost function of unit n (\$)
M	constant used for integer modeling
$P_{(n)}^{Min}$	minimum power generation limit of unit n (MW)
$P_{(n)}^{Max}$	maximum power generation limit of unit n (MW)
$P_{(n)}^{RU}$	ramp-up power generation limit of unit n (MW/h)
$P_{(n)}^{RD}$	ramp-down power generation limit of unit n (MW/h)
$PE_{(d)}^{WDU}$	electric energy consumption of water desalination unit d (kW/m3)
$PL_{(i,j)}^{Max}$	capacity of the transmission lines between buses i and j (MW)
$PR_{(s)}^{BSS}$	power rating of the battery storage s (MW)
$PT_{(d)}^{WDU}$	thermal energy consumption of water desalination unit d (MJ/m3)
$T_{(n)}^{Up}$	minimum up-time limit of unit n (h)
$T_{(n)}^{Down}$	minimum down-time limit of unit n (h)
$WC_{(d)}^{Tank}$	capacity of the storage tank for water desalination unit d (m3)
$Wd_{(d,h)}^{WDU}$	hourly water demand of water desalination unit d and at time period h (m3)
$Wd_{(d)}^{Peak}$	daily peak water demand of water desalination unit d (m3)
$Wd_{(d,h)}^{Profile}$	normalized water demand of water desalination unit d and at time period h (m3)
$Wg_{(d)}^{TDC}$	total daily generation capacity of water desalination unit d (m3)
$WPC_{(d)}^{Cha}$	water charging capacity of the storage tank for water desalination unit d (m3)
$WPC_{(d)}^{Dis}$	water discharging capacity of the storage tank for water desalination unit d (m3)
$\lambda_{m,n}$	slope of the power generation cost curve for segment m of unit n (\$/MW)

Binary variables

$BB_{(s,h)}^{Cha}$	power charging indicator binary variable of battery
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	storage s and at period h (MW)
$BB_{(s,h)}^{Dis}$	power discharging indicator binary variable of battery storage s and at period h (MW)
$BG_{n,h}$	binary variable indicating on/off status of plant n at time period h
$BT_{(d,h)}^{Cha}$	charging indicator binary variable of storage tank of desalination unit d at time period h (m3)
$BT_{(d,h)}^{Dis}$	discharging indicator binary variable of storage tank of desalination unit d at time period h (m3)

Continuous variables

$CG_{(n,h)}^{SU}$	start-up cost of plant n at time period t (\$)
$CG_{(n,h)}^{SD}$	shut-down cost of plant n at time period t (\$)
$CG_{(n,h)}^{PG}$	power generation cost of plant n at time period h (\$)
$PB_{(i,h)}^{Net}$	net power generated by all batteries located at bus i and at time period h (MW)
$PB_{(s,h)}^{Dis}$	charged power to the battery storage s and at time period h (MW)
$PB_{(s,h)}^{Cha}$	discharged power to the battery storage s and at time period h (MW)
$PD_{(i,h)}^{Net}$	net power consumed by all desalination units located at bus i and at time period h (MW)
$Pg_{(s,h)}^{BSS}$	net power generated by battery storage s and at time period h (MW)
$PG_{(i,h)}^{Net}$	net power generated by all generators located at bus i and at time period h (MW)
$PL_{(i,j,h)}$	power flow of transmission lines between buses i and j and at time period h (MW)
$PL_{(i,h)Net}$	net power injection to the all lines connected to bus i and time period h (MW)
$PLD_{(i,h)}^{Bus}$	electric power demand at bus i and time period h (MW)
$TB_{(s,h)}^{Cha}$	charging power time duration of battery storage s at time period h (h)
$TB_{(s,h)}^{Dis}$	discharging power time duration of battery storage s at time period h (h)
TC	total daily operation cost (\$)
$TW_{(d,h)}^{Cha}$	charging water time duration of storage tank of desalination unit d at time period h (h)
$TW_{(d,h)}^{Dis}$	discharging water time duration of storage tank of desalination unit d at time period h (h)
$W_{(d,h)}^{Cha}$	charged water to the storage tank of desalination unit d and at time period h (m3)
$W_{(d,h)}^{Dis}$	discharged water from the storage tank of desalination unit d and at time period h (m3)
$W_{(d,h)}^{Tank}$	water stored in the storage tank of desalination unit d and at time period h (m3)
$Wg_{(d,h)}^{Tank}$	net water generated by storage tank of desalination unit d and at time period h (m3)
$Wg_{(d,h)}^{WDU}$	water generated by desalination unit d and at time period h (m3)
$Wg_{(d)}^{HC}$	maximum hourly water generation capacity of desalination unit d (m3/h)
$\delta_{(i,h)}$	voltage angle of bus i and time period h (rad)
$\delta P_{m,n,h}$	power generated at segment m of unit n at time period h (MW)
$\Delta P_{m,n}$	power generation limit of segment m of unit n at time period h (MW)

which is, in turn, composed of the three components, power generation (fuel), start-up, and shut-down costs. Until now, various mathematical UC models have been proposed and used in practice [14–17].

At present, the energy consumption of the desalination units is

considered as a constant value equal to its nominal capacity in the unit commitment problem. In other words, the hourly planning of these units is not considered in the UC problem. By simultaneously scheduling these units with thermal units in the unit commitment problem,

better results can be achieved in terms of operation costs. The results of the conducted in [18] validate this claim. This is due to the high energy consumption of desalination plants and their effective role in the problem results. This is especially important when the desalination unit also has an internal water storage tank. Because by scheduling and controlling this resource concurrently with other control variables in the network, the scheduling objectives can more effectively be achieved [18]. Today, grid-scale storage batteries have become an integral part of the power grids. Installation of these batteries in the network has brought significant growth over the past years [19]. One of their most important applications is energy consumption management with load profile leveling [20]. To date, the research conducted in [18] is the only study on the simultaneous scheduling of desalination units within the UC problem. In this study, a general model of desalination is considered and its technology (membrane or thermal) was not considered. In addition, the effect of any of the above factors is neither modeled nor taken into account. In other words, the significant effect of water storage tank inside unit as well as storage batteries integrated in the transmission network is not considered. Accordingly, this paper presents a new model for simultaneous short-term scheduling of water desalination units and thermal power plants. The proposed model has modeled and incorporated various types of membrane and thermal desalination technologies. In addition, the internal water storage tank of each desalination unit and storage batteries installed in the transmission network are modeled and their charge and discharge scheduling optimally determined simultaneously with the desalination and thermal units. All of the above is done by modeling and considering the constraints of the power transmission grid. The final proposed model is a Mixed Integer Linear Programming (MILP) optimization model does not have divergence problems and can be easily solved by existing commercial software without concerns about size. In summary, the novelties of this paper can be listed as follows.

- 1.– Modeling and integration of water desalination units in thermal unit commitment problem.
- 2.– Various membrane and thermal desalination process are considered.
- 3.– Internal water storage tank of desalination units is considered.
- 4.– Battery energy storage system integrated in the transmission network is considered.
- 5.– Power transmission network constraints are considered.

2. The proposed model

The aim of the proposed model is to optimally schedule a system

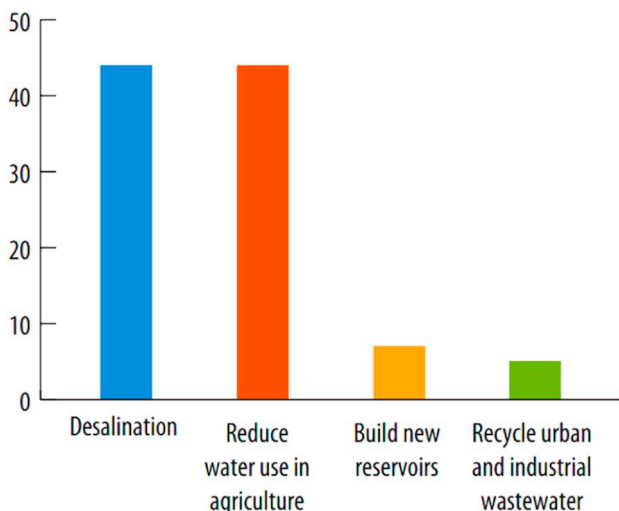


Fig. 1. Source of new water supplied by 2050 (percent) [4].

composed of a set of the thermal power plants (TPPs) supplying load demands of an electric power transmission network. The load demand is composed of the various Water Desalination Units (WDUs) in addition to the local electric demand of the consumers connected to the network buses. Each WDU supplies dedicated local water demand independently and there is not a link between the WDUs except the electric transmission grid. The mean of scheduling is optimally defining three decision variables. The first one is to determine optimal on/off status and also generated power by the TPPs considering thermal inertia. The second and third ones are the decision variables related to the WDUs and Battery Energy Storage Systems (BESSs), which will be introduced in the following. Final goal of the scheduling is to find the minimum cost operation scheduling in which all load demand will be completely supplied. The system is also equipped with the large-scale BESSs. Role of the batteries is to level the load profile by peak shaving followed by the valley filling. Each block of the proposed model is explained and formulated in the following.

2.1. Water desalination units

Fig. 2 depicts schematic of a generic WDU equipped with water storage tank. As in the figure, the WDU is supplied from the two sources. The first one is the input saline water. Salinity content may significantly vary based on the source of water. Two main types of source water are seawater and brackish water. Other input to the WDU is the required energy for the process in addition to the auxiliary systems used [21]. The input energy may be in the form of electrical and/or thermal based on the main process. The Desalination processes can be broadly divided to the membranes and thermal process. The Reverse Osmosis (RO) method which is the most known desalination method belongs to the membrane processes category. The RO method used only electric energy to produce potable water. On the other hand, MSF and MED are thermal technologies which consume both of the electric and thermal energies in the process. Therefore, type and level of the required energy is a function of the WDU main process, namely RO, MSF, or MED. Other factors such as input water quality and temperature, output water quality, distance from the source water and demand location, and so on may affect energy demand [22]. However, in a specific project location these factors are the same and do not change.

Output of the WDU is the freshwater in addition to the high saline content or brine which will be fed back to the source of the water. The produced freshwater will be divided to two streams, as illustrated in the figure. The first one directly supplies the water demand while the another one charges the water storage tank. The water stored in the tank will be used at other time periods when needed. The water storage tank helps to leveling WDU loading and lessens its stress by shaving the demand peak [23].

The water injection into the tank and also withdrawal from it will be performed by utilizing two independent electric motors, namely charging and discharging water-pumps. The WDU in addition to the output water of the tank supply the local water demand simultaneously. The hourly generated water by the WDU, charging water, discharging water, and also stored water in the tank should be determined in an optimal fashion to achieve a minimum cost daily scheduling. To do this, a mathematical model must be introduced. In Eq. (1) balance between generated and consumed water in each WDU equipped with water storage tank is established. As in the equality, the net water generated by the tank in addition to the water supplied by the process is equal to the water demand at any time period of the operation for each WDU. The hourly water demand for each WDU is itself a function of the peak water demand and the hourly demand profile, as formulated in Eq. (2). Each WDU demand required energy based on the process, as explained above. The total equivalent demanded electric energy is therefore a function of the unit electric and thermal energy needed multiplied by the water generated, as in Eq. (3). The equivalent electric energy demand is calculated by adding required electric energy and equalized

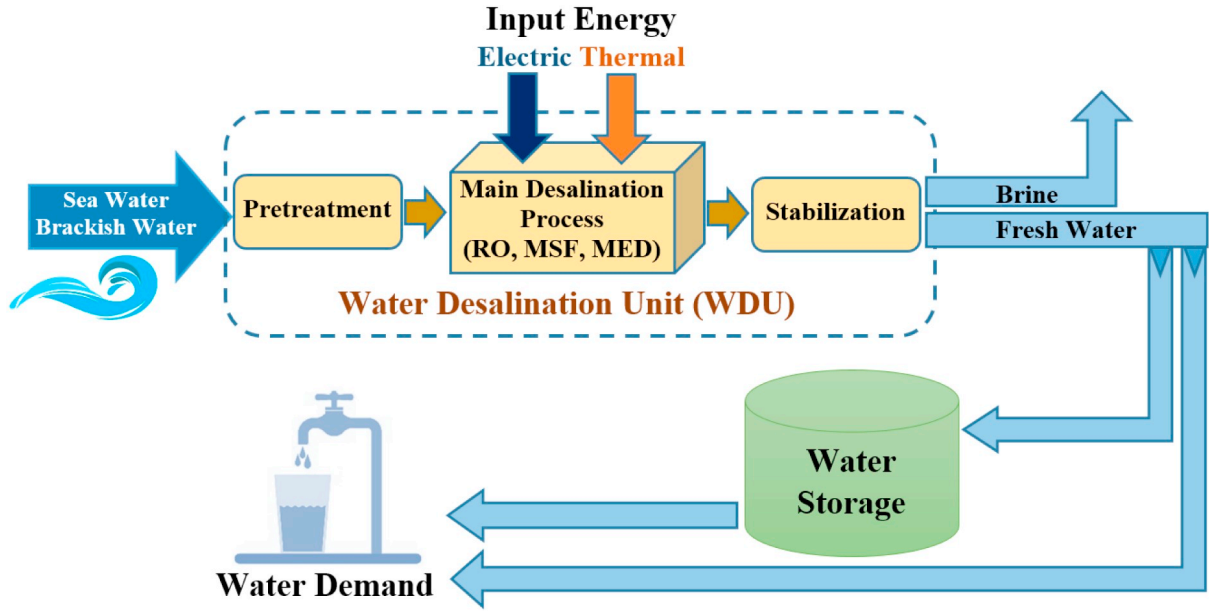


Fig. 2. Schematic of a generic WDU with water storage tank.

thermal energy needed. The electric equivalent of the thermal energy is calculated by using a conversion factor to move from the mega-joules to kilowatt-hours. The final energy demand is divided to 1000 to be consistent with the dimension of the local electric demand of the network buses which is in Megawatt. The water generated by the WDU at any time period is a positive value and cannot exceed the hourly capacity of the water generation. This is mathematically declared in Eq. (4). The hourly water generation capacity is itself a function of the total daily capacity of the WDU, as in Eq. (5).

$$Wg_{(d,h)}^{Tank} + Wg_{(d,h)}^{WDU} = Wd_{(d,h)}^{WDU} \quad \forall d \in \Omega_D, h \in \Omega_H \quad (1)$$

$$Wd_{(d,h)}^{WDU} = Wd_{(d)}^{Peak} Wd_{(d,h)}^{Profile} \quad \forall d \in \Omega_D, h \in \Omega_H \quad (2)$$

$$Pd_{(d,h)}^{WDU} = \frac{Wg_{(d,h)}^{WDU}}{1000} (PE_{(d)}^{WDU} + K_{Joule}^{Wh} PT_{(d)}^{WDU}) \quad \forall d \in \Omega_D, h \in \Omega_H \quad (3)$$

$$0 \leq Wg_{(d,h)}^{Tank} \leq Wg_{(d)}^{HC} \quad \forall d \in \Omega_D, h \in \Omega_H \quad (4)$$

$$Wg_{(d)}^{HC} = Wg_{(d)}^{TDC} / 24 \quad \forall d \in \Omega_D, h \in \Omega_H \quad (5)$$

The net water supplied by the water storage tank is calculated in Eq. (6) and is equal to the difference between charged water to and discharged water from the tank. Also, the hourly charging and discharging waters are positive values and cannot go beyond the dedicated water-pump capacity. Inequalities formulated in Eqs. (7) and (8) enforce this limitation on the charging and discharging waters of the tank, respectively. For the sake of energy saving purposes, the tank cannot be simultaneously charged and discharged at a specific time period. In other word, only one water-pump can work at any specific time period. This is mathematically formulated by using two binary variables. Each one indicates one of the charging and discharging states which their binary nature is denoted by Eqs. (9) and (10). In Eq. (11), the problem is enforced to switch-on only one of the binary indicators and therefore perform one action. Consequently, the tank can be charged or discharged if the corresponding binary variable is switched-on previously, as formulated in Eqs. (12) and (13). The large enough constant M is used to not limit the upper bounds of the charging and discharging variables to number of one because they are previously bounded in Eqs. (7) and (8).

$$Wg_{(d,h)}^{Tank} = Wd_{(d,h)}^{Dis} - Wd_{(d,h)}^{Cha} \quad \forall d \in \Omega_D, h \in \Omega_H \quad (6)$$

$$0 \leq Wd_{(d,h)}^{Cha} \leq WPC_{(d)}^{Cha} \quad \forall d \in \Omega_D, h \in \Omega_H \quad (7)$$

$$0 \leq Wd_{(d,h)}^{Dis} \leq WPC_{(d)}^{Dis} \quad \forall d \in \Omega_D, h \in \Omega_H \quad (8)$$

$$BT_{(d,h)}^{Cha} \in \{0, 1\} \quad \forall d \in \Omega_D, h \in \Omega_H \quad (9)$$

$$BT_{(d,h)}^{Dis} \in \{0, 1\} \quad \forall d \in \Omega_D, h \in \Omega_H \quad (10)$$

$$BT_{(d,h)}^{Cha} + BT_{(d,h)}^{Dis} \leq 1 \quad \forall d \in \Omega_D, h \in \Omega_H \quad (11)$$

$$Wd_{(d,h)}^{Cha} \leq BT_{(d,h)}^{Cha} M \quad \forall d \in \Omega_D, h \in \Omega_H \quad (12)$$

$$Wd_{(d,h)}^{Dis} \leq BT_{(d,h)}^{Dis} M \quad \forall d \in \Omega_D, h \in \Omega_H \quad (13)$$

The water stored in the tank at any time period is a linear function of the water remained from the previous time period in addition to the net water interchange at the present time period, as it is formulated in Eq. (12). In Eq. (13), it is stated that the volume of water stored in the tank is a positive variable and should be limited to the tank capacity. Furthermore, volume of the discharged water from the tank at any time period cannot go over the remained water in the tank from the previous time period, as denoted by Eq. (14). Also, the water stored in each tank cannot exceed its capacity, as denoted by Eq. (15). Additionally, volume of the water discharged from each tank should be less than the maximum stored water, as in Eq. (16). Finally, a daily water balance for the tank is considered. In other word, the tank ends the time periods with the same water it starts with. This is because of the health concerns related to the duration of the water storage in the tank. Equality expressed in Eq. (17) denotes this consideration mathematically.

$$Wd_{(d,h)}^{Tank} = Wd_{(d,h-1)}^{Tank} + Wd_{(d,h)}^{Cha} TW_{(d,h)}^{Cha} - Wd_{(d,h)}^{Dis} TW_{(d,h)}^{Dis} \quad \forall d \in \Omega_D, h \in \Omega_H \quad (14)$$

$$0 \leq Wd_{(d,h)}^{Tank} \leq WC_{(d)}^{Tank} \quad \forall d \in \Omega_D, h \in \Omega_H \quad (15)$$

$$Wd_{(d,h)}^{Dis} TW_{(d,h)}^{Dis} \leq Wd_{(d,h-1)}^{Tank} \quad \forall d \in \Omega_D, h \in \Omega_H \quad (16)$$

$$\sum_h (Wd_{(d,h)}^{Cha} TW_{(d,h)}^{Cha}) = \sum_h (Wd_{(d,h)}^{Dis} TW_{(d,h)}^{Dis}) \quad \forall d \in \Omega_D \quad (17)$$

2.2. Battery energy storage systems

Fig. 3 illustrates a typical BESS connected to one of the network

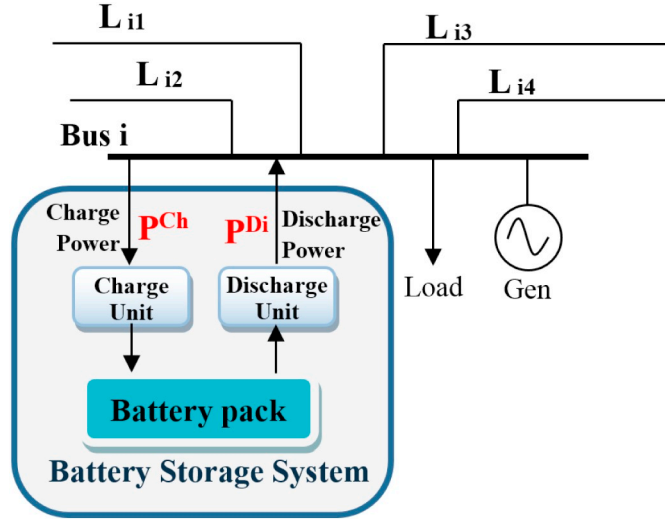


Fig. 3. Schematic of a generic BESS and its connection impact [24].

buses [24]. As in can be observed from the figure, the BESS interacts with the bus, and consequently the grid, by injecting discharge or drawing charge power. The whole BESS therefore can be tabulated as a combination of a fictitious load in addition to a fictitious generator indicating charging and discharging states, respectively [25].

A grid-scale BESS is generally composed of various parts. The first is heart of the system called the battery container. The battery container itself consists of battery packs. Each battery pack itself consists of battery modules, which are also made up of battery cells. In other words, the battery container is a collection of connecting a large number of battery cells. The main function of the whole battery storage system which is to store and discharge electrical energy will be performed in these cells. It should be noted that the cells do this on direct current (DC) electricity. In the grid-scale applications, wherein the battery interacts with the main AC utility grid, a bidirectional power converter is used between the battery container and the grid. The power converter changes AC power of the grid to the DC one (rectifier conversion mode) at the charging state of the battery. On the contrary, the power converter works at the inverter mode by converting DC stored energy in the battery to the grid AC power when then battery is discharging. If, nominal voltage of the point of connection be higher than the output voltage of the converter, a (set of) transformer(s) will be used to match the voltages. Usually, high voltage connection of the battery needs a transformer between the converter and the grid. Therefore, at the high voltage grid-scale connections the whole battery energy storage system comprises at least three main parts, namely battery container, power converter, and transformer. Other auxiliary equipment is also used with the battery system for protection, control, and monitoring purposes. As the above concise survey presents, the BESS composed of various parts which are often expensive. This has made the BESS costlier compared to the other conventional network equipment. Although the operation and maintenance (O&M) cost the batteries is very low, the very high initial investment cost has been a major obstacle to their extensive use. This issues highlights the importance of adopting a proper operation schedule in order to maximizing the BESS benefits.

The net power generated by the BESS is the difference between injected (discharged) power and drawn (charged) one, as in Eq. (18). This net power may be negative or positive depend on the BESS status. Anyway, this value should not be greater than the power rating of the BESS. This limitation is considered in Eq. (19). On the other hand, stored energy is a positive value and lower than the energy capacity of the battery pack within the whole BESS system, as denoted by Eq. (20). Eq. (21) states that energy stored in the BESS at any time period is the

summation of previously stored energy and net power exchange occurred. Similar to the water storage, two binary indicators are used to model charging and discharging states, as denoted by Eqs. (22) and (23). Inequalities shown in Eq. (24) to Eq. (26) enforce the BESS to select only one of the charging and discharging actions at any time period of operation by using two indicator binary variables. Discharged energy from the BESS at any time period should be equal to or lower than the whole energy stored previously considering discharging efficiency, as stated by Eq. (23). Finally, energy balance of the BESS during daily operation is established in Eq. (24).

$$P_{g(b,h)}^{BSS} = PB_{(b,h)}^{Dis} - PB_{(b,h)}^{Cha} \quad \forall b \in \Omega_B, h \in \Omega_H \quad (18)$$

$$-PR_{(b)}^{BSS} \leq P_{g(b,h)}^{BSS} \leq +PR_{(b)}^{BSS} \quad \forall b \in \Omega_B, h \in \Omega_H \quad (19)$$

$$0 \leq E_{(b,h)}^{BSS} \leq ER_{(b)}^{BSS} \quad \forall b \in \Omega_B, h \in \Omega_H \quad (20)$$

$$E_{(b,h)}^{BSS} = E_{(b,h-1)}^{BSS} + PB_{(b,h)}^{Cha} TB_{(b,h)}^{Cha} EB_{(b)}^{Cha} - PB_{(b,h)}^{Dis} TB_{(b,h)}^{Dis} / EB_{(b)}^{Dis} \quad \forall b \in \Omega_B, h \in \Omega_H \quad (21)$$

$$BB_{(b,h)}^{Cha} \in \{0, 1\} \quad \forall b \in \Omega_B, h \in \Omega_H \quad (22)$$

$$BB_{(b,h)}^{Dis} \in \{0, 1\} \quad \forall b \in \Omega_B, h \in \Omega_H \quad (23)$$

$$BB_{(b,h)}^{Cha} + BB_{(b,h)}^{Dis} \leq 1 \quad \forall b \in \Omega_B, h \in \Omega_H \quad (24)$$

$$0 \leq PB_{(b,h)}^{Cha} \leq BB_{(b,h)}^{Cha} M \quad \forall b \in \Omega_B, h \in \Omega_H \quad (25)$$

$$0 \leq PB_{(b,h)}^{Dis} \leq BB_{(b,h)}^{Dis} M \quad \forall b \in \Omega_B, h \in \Omega_H \quad (26)$$

$$PB_{(b,h)}^{Dis} TB_{(b,h)}^{Dis} \leq E_{(b,h-1)}^{BSS} EB_{(b)}^{Dis} \quad \forall b \in \Omega_B, h \in \Omega_H \quad (27)$$

$$\sum_h (PB_{(b,h)}^{Cha} TB_{(b,h)}^{Cha} EB_{(b)}^{Cha}) = \sum_h (PB_{(b,h)}^{Dis} TB_{(b,h)}^{Dis} / EB_{(b)}^{Dis}) \quad \forall b \in \Omega_B \quad (28)$$

2.2.1. Thermal power plants

The thermal power plants (TPP) convert the potential energy of the fossil fuels to the electric energy in a thermal cycle. In the problems concerning with the economic operation of power systems, the input-output set of a thermal power plant is of fundamentals factors. In the simplest form, there is a boiler-turbine-generator set. This set includes a boiler that produces steam to power a turbine-generator couple. The electricity produced by the generator is supplying both the utility grid and also auxiliary power systems in the power plant itself. The latter, including boiler feed pumps, fans, condenser circulating water pumps and so on, may require 2–6% of the gross output of the unit in atypical plant. While defining the unit characteristics, the gross input versus net output will be considered. The net output of the plant is the electrical power injected to the utility grid. Fig. 4 illustrates the input-output curve of a steam unit in idealized form. The input to the unit shown on the ordinate may be either in terms of heat energy requirements in Million British thermal unit (MBtu) per hour or in terms of total cost per hour. In the second case, MBtu per hour heat input is multiplied by the unit cost of fuel cost in dollars per MBtu per hour. The resulting unit therefore will be dollars per hour as in the figure. The output is typically the net electrical output of the unit irrespective of the self-supply requirements. These data constructing the curve may be attained from design calculations or from heat rate field tests. When heat rate experimental figures are utilized, the data used often do not form a smooth curve. Here, an idealized curve is represented for the purpose of operation planning studies [13,26].

For the purpose of economic operation scheduling, a piecewise linear approximation of this non-linear function can be used to avoid nonlinearities [27]. In this regard, original quadratic curve will be divided to a set of straight lines with first order polynomial definition. Accuracy of the approximation can be enhanced by increasing number of the straight line but at the expense of the run time. It should be noted

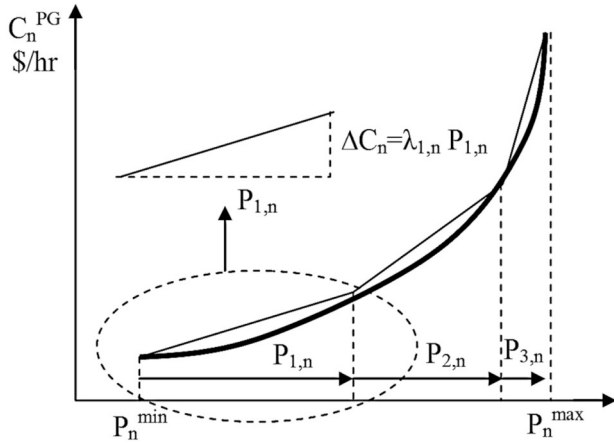


Fig. 4. Input-output curve of a steam plant and piecewise linearization technique.

that, a binary indicator variable is used to model on/off states of each unit at any time period. This is mathematically expressed by Eq. (29). Approximated linear cost function is mathematically expressed in Eq. (30). Each power block is constrained to the corresponding maximum value, as in Eq. (31). More details can be found in [28]. By using this approximation, total generated power by each unit at any time period will be equal to Eq. (32). This value should be within the upper and lower levels of the unit power generation capacity, as denoted by Eq. (33).

$$BG_{n,h} \in \{0, 1\} \quad \forall b \in \Omega_B, \quad h \in \Omega_H \quad (29)$$

$$CG_{(n,h)}^{PG} = k_n \times BG_{n,h} + \sum_m (\lambda_{m,n} \times \delta P_{m,n,h}) \quad n \in \Omega_N, \quad h \in \Omega_H \quad (30)$$

$$0 \leq \delta P_{m,n,h} \leq \Delta P_{m,n} \quad n \in \Omega_N, \quad h \in \Omega_H \quad (31)$$

$$P_{g(n,h)}^{Gen} = P_{(n)}^{Min} \times BG_{n,h} + \sum_m \delta P_{m,n,h} \quad \forall n \in \Omega_N, \quad h \in \Omega_H \quad (32)$$

$$BG_{n,h} P_{(n)}^{Min} \leq P_{g(n,h)}^{Gen} \leq BG_{n,h} P_{(n)}^{Max} \quad \forall n \in \Omega_N, \quad h \in \Omega_H \quad (33)$$

Besides power generation, each unit has its own start-up and shut-down cost related to the operation status. Each time the unit is started-up, it requires fuel and therefore costs to warm it up. And similarly, shutting-down the unit every time means paying on energy. As a result, unit start-up and shut-down costs at each time interval are subject to its on/off status at the previous period. These terms are mathematically modeled in Eqs. (34)–(35) and (36)–(37) for the startup and shutdown costs, respectively.

$$CG_{(n,h)}^{SU} = C_{(n)}^{On} (BG_{n,h} - BG_{n,h-1}) \quad \forall n \in \Omega_N, \quad h \in \Omega_H \quad (34)$$

$$CG_{(n,h)}^{SU} \geq 0 \quad \forall n \in \Omega_N, \quad h \in \Omega_H \quad (35)$$

$$CG_{(n,h)}^{SD} = C_{(n)}^{Off} (BG_{n,h-1} - BG_{n,h}) \quad \forall n \in \Omega_N, \quad h \in \Omega_H \quad (36)$$

$$CG_{(n,h)}^{SD} \geq 0 \quad \forall n \in \Omega_N, \quad h \in \Omega_H \quad (37)$$

Due to the thermal dynamics, the generated power by the unit cannot quickly change. In other word, power generation adjustment at any time period has a limitation lower than the rated capacity of the unit. This limitation for the increase and decrease in the generated power is known as ramp-up and ramp-down, respectively. Each unit has its own values for ramp-up and ramp-down related to the thermal dynamics. Power generation ramp-up and ramp-down are modeled in Eqs. (38) to (41). More details can be found in [28].

$$\sum_{t=k}^{k+T_{(n)}^{Up}-1} BG_{n,t} \geq T_{(n)}^{Up} (BG_{n,h} - BG_{n,h-1}) \quad \forall n \in \Omega_N, \quad h = 1, \dots, H - T_{(n)}^{Up} + 1 \quad (38)$$

$$\sum_{t=k}^H [BG_{n,t} - (BG_{n,h} - BG_{n,h-1})] \geq 0 \quad \forall n \in \Omega_N, \quad h = H - T_{(n)}^{Up} + 2, \dots, H \quad (39)$$

$$\sum_{t=k}^{k+T_{(n)}^{Down}-1} (1 - BG_{n,t}) \geq T_{(n)}^{Down} (1 - BG_{n,h} - BG_{n,h-1}) \quad \forall n \in \Omega_N, \quad h = 1, \dots, H - T_{(n)}^{Down} + 1 \quad (40)$$

$$\sum_{t=k}^H [1 - BG_{n,t} - (1 - BG_{n,h} - BG_{n,h-1})] \geq 0 \quad \forall n \in \Omega_N, \quad h = H - T_{(n)}^{Down} + 2, \dots, H \quad (41)$$

Finally, each thermal unit must inevitably stay on its status (on or off) for some times and then be allowed to change status. In other words, the minimum time duration of on or off status of the unit cannot be less than the specified amount known to the minimum up-time and minimum down-time respectively. These constraints are also modeled in Eqs. (42) and (43) [29].

$$P_{g(n,h)}^{Gen} - P_{g(n,h-1)}^{Gen} \leq P_{(n)}^{RU} [1 - (BG_{n,h} - BG_{n,h-1})] \quad \forall n \in \Omega_N, \quad h \in \Omega_H \\ + P_{(n)}^{Min} (BG_{n,h} - BG_{n,h-1}) + P_{(n)}^{Max} (1 - BG_{n,h}) \quad (42)$$

$$P_{g(n,h-1)}^{Gen} - P_{g(n,h)}^{Gen} \leq P_{(n)}^{RD} [1 - (BG_{n,h-1} - BG_{n,h})] \quad \forall n \in \Omega_N, \quad h \in \Omega_H \\ + P_{(n)}^{Min} (BG_{n,h-1} - BG_{n,h}) + P_{(n)}^{Max} (1 - BG_{n,h-1}) \quad (43)$$

It should be noted that, in practice some of the wasted heat of the TPPs can be recovered. The recovered heat employs to supply thermal energy needed in the thermal-based desalination technologies. This can be performed if the TPP and also WDU are located in the same location. In this situation, a certain percentage of the output energy of the TPPs can be regarded as recovered heat fed into the WDU. In other word, in the modeling of the TPP, besides main output electrical power, a portion will be considered for recovered heat. Summation over the both electrical and also recovered heat of the TPP should be equal to the total output energy of the unit.

2.3. Transmission network (lines and buses)

The transmission grid is the corridor between various generators and consumers. Each equipment is connected to a specific bus of the transmission network. The buses are connected together by means of transmission lines and constitute a mesh grid. Fig. 5 presents a network bus with connected lines and equipment [30]. The equipment may be power producer like generators, power consumer like local demands and WDUs, or interactive like the BESSs.

Balance of the total generated power with the total consumed power should be kept in the bus at any time period of operation. This is mathematically established in Eq. (44). As in the equation, right hand side items are injections to the bus while left hand side ones are withdrawals. The injections are power produced by the connected generators in addition to the net injection of the BESSs. It should be noted that as it was explained previously, the latter item is not necessarily positive. On the other hand, the powers leaving the bus are local bus demand, local WDUs, and net injected power to the lines connected to the bus.

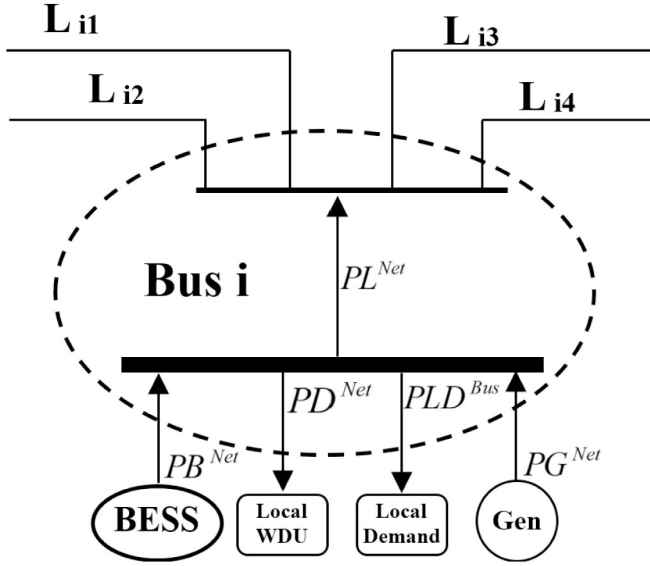


Fig. 5. Representation of a network bus with connected lines and equipment.

$$PG_{(i,h)}^{Net} + PB_{(i,h)}^{Net} = PLD_{(i,h)}^{Bus} + PL_{(i,h)}^{Net} + PD_{(i,h)}^{Net} \quad \forall i \in \Omega_I, h \in \Omega_H \quad (44)$$

It should be noted that net generated power at the bus is equal to the summation over power generated by all generators connected to the bus, as denoted by Eq. (45). There is a similar situation for the other equipment connected to the bus. In other word, equations Eq. (46) to Eq. (48) calculate net power injected by the BESS, net power extracted by the WDUs, and net power injected to the connected lines, respectively. The flow of the power in a transmission lines is a linear function of the difference between voltage angles of the buses located at the beginning and end of the line, as modeled in Eq. (49). Finally, line power flow should be lower than the line capacity which is denoted by Eq. (50).

$$PG_{(i,h)}^{Net} = \sum_{n \in \Omega_{NB}} PG_{(n,h)}^{Gen} \quad \forall i \in \Omega_I, h \in \Omega_H \quad (45)$$

$$PB_{(i,h)}^{Net} = \sum_{s \in \Omega_{SB}} PG_{(s,h)}^{BSS} \quad \forall i \in \Omega_I, h \in \Omega_H \quad (46)$$

$$PD_{(i,h)}^{Net} = \sum_{d \in \Omega_{DB}} PD_{(d,h)}^{WDU} \quad \forall j \in \Omega_I, h \in \Omega_H \quad (47)$$

$$PL_{(i,h)}^{Net} = \sum_j PL_{(i,j,h)} \quad \forall i, j \in \Omega_I, h \in \Omega_H \quad (48)$$

$$PL_{(i,j,h)} = \frac{\delta_{(i,h)} - \delta_{(j,h)}}{BL_{(i,j)}} \quad \forall i, j \in \Omega_I, h \in \Omega_H \quad (49)$$

$$-PL_{(i,j)}^{Max} \leq PL_{(i,j,h)} \leq PL_{(i,j)}^{Max} \quad \forall i, j \in \Omega_I, h \in \Omega_H \quad (50)$$

2.4. Objective function and components

The objective function of the problem is to minimize daily operation cost. The operation cost is composed of three cost components including power generation (fuel) cost, start-up cost, and shut-down cost of the thermal power plants. In Eq. (51), these cost components are summed up and elaborated over all time periods and generating units to calculate total daily operation cost. This value should be minimized by controlling decision variables of the problem. The decision variables can be mainly categorized as thermal power plants variables, the BESSs variables, and the WDUs variables. The TPPs variables are hourly on/off status in addition to the hourly power generation. For the BESSs the variables consist of hourly charging or discharging powers and

consequently stored energy. Finally, the variables of the WDU which should be determined optimally are hourly water generation in addition to the hourly water charging or discharging of the tank.

$$\begin{aligned} \text{Min} \quad & TC = \sum_n \sum_h (CG_{n,h}^{PG} + CG_{n,h}^{SU} + CG_{n,h}^{SD}) \\ & W_{(d,h)}^{WDU}, W_{(d,h)}^{Dis}, W_{(d,h)}^{Cha} \\ & PB_{(b,h)}^{Cha}, PB_{(b,h)}^{Dis}, PG_{(n,h)}^{Gen} \end{aligned} \quad (51)$$

3. Case study

In order to evaluate efficiency and functionality, the model proposed in the previous section is implemented on a test case. The 24 bus network of IEEE Reliability test system (RTS) is used as the test system [31]. The system has 32 generation units along with 35 transmission lines. The data for generating units are provided in Appendix while the data for the lines can be found in [31]. The synchronous condenser in the original system which was installed in bus 14 is disconnected because of ignoring reactive power in the DC power flow. All of the generating units are as the original data but with some modifications as follows. In the original system, on/off status of the generating units is not defined. For the purpose of unit commitment study, it is supposed that units 9–11 and 20–32 are initially online. Also, shut-down cost and start-up cost of each unit is set to 1500\$. The considered operation horizon is composed of 24 slices each one with one-hour duration. Schematic of the utility power grid in addition to the BESSs and WDUs connection buses is depicted at Fig. 6.

Two grid-scale battery storage systems (BESS) and four water desalination units (WDU) with different capacities are added to the system. The batteries, abbreviated by BESS1 and BESS2, are connected to the buses 10 and 19 of the network. Detailed data of the batteries is given in Table 1. The desalination units, abbreviated by WDU1 to WDU4, are located in buses 4, 8, 17, and 20 of the network. Various

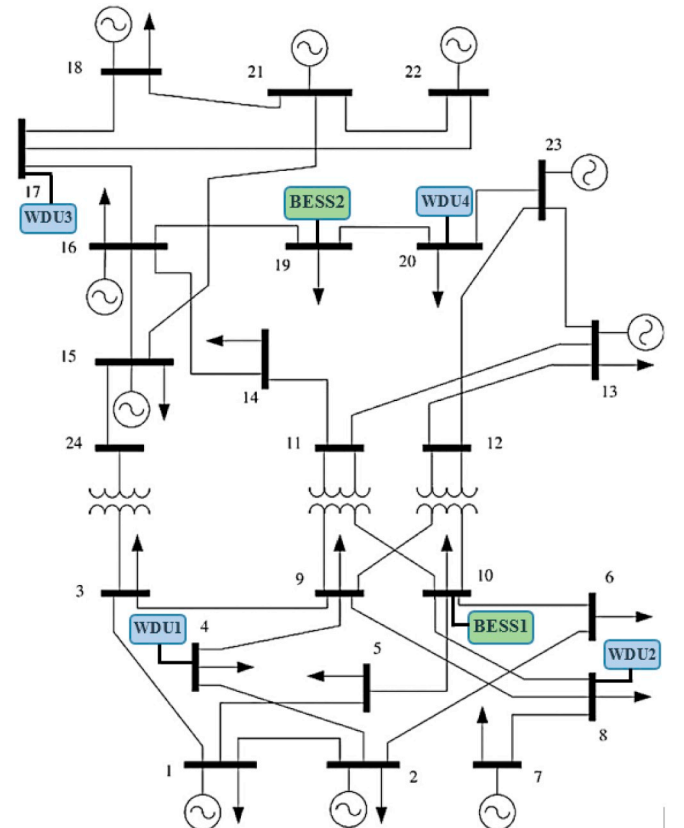


Fig. 6. Diagram of the IEEE 24-bus RTS [33].

Table 1
Data of the battery storage system.

Title	Symbol	Unit	BESS1	BESS2
Bus location	–	–	10	19
Technology	–	–	Li-ion	Lead-acid
Power rating	$PR_{(s)}^{BSS}$	MW	60	80
Energy capacity	$ER_{(s)}^{BSS}$	MWh	240	480
Efficiency	$EB_{(s)}$	%	0.85	0.9

characteristics of the WDUs are shown in Table 2, namely bus location, technology, feed water, recovery ratio, energy consumption, capacity, daily peak water demand, and data of the water storage tank. Each desalination unit is equipped with an internal water storage tank and supplies associated local water demand individually.

For the sake of simplicity and without loss of generality, it is supposed that hourly water demand for all of the WDU units follows a similar profile but with different hourly values. Profile of the water demand for all desalination units, which is the same among the WDUs, is depicted in Fig. 7. This profile will be multiplied by the local daily peak water demand of each unit (provided in Table 2) to yield hourly water demand of the day. Finally, hourly electric power demand of the system is depicted in Fig. 8 with the peak value of 2850 MW. The final MILP model is implemented in the GAMS software [32] and solved using CPLEX software [33]. Two cases are simulated entitled Base Case and Double Storage. The Base Case refers to the case wherein there is neither battery storage nor water storage in the system. Accordingly, the Double Storage case denotes the case with both storage system, i.e., battery storage and water storage. The models are simulated on a laptop computer with a Core i5, 2.5 GHz CPU and with 4.00 GB of memory. Runtime for the min model, namely double storage, was about 9 s.

General cost results of the simulations are given in Table 3. The table includes total start-up and shut-down cost in addition to the total generation cost of the generating units for the both cases. In addition, net and relative difference of the values are reported. As the results show, all components of the total operation cost have been experienced a reduction after employing storage systems. Strictly speaking, the start-up cost shows the highest reduction with slightly more than 26%. Also, total shut-down cost of the units is reduced from 3060 to 2280 which is equal to 780 dollars or equivalently about 25.5%. The largest portion of the total operation cost is related to the power generation cost which, in turn, is because of the fuel consumption of the generating units. This cost term demonstrates a 4.694% reduction. Considering high share of this value, the total cost is also influenced. As in the table, by employing both of the electric energy storage and water storage in the tanks, 49,978 dollars of the daily operation cost will be saved which is equal to more than 5%. This is a considerable saving which will be repeated each day with a more or less same value with respect to the system operation status.

Utilizing both types of the storage in the system in addition to reducing the cost of daily operation has several other benefits, each of

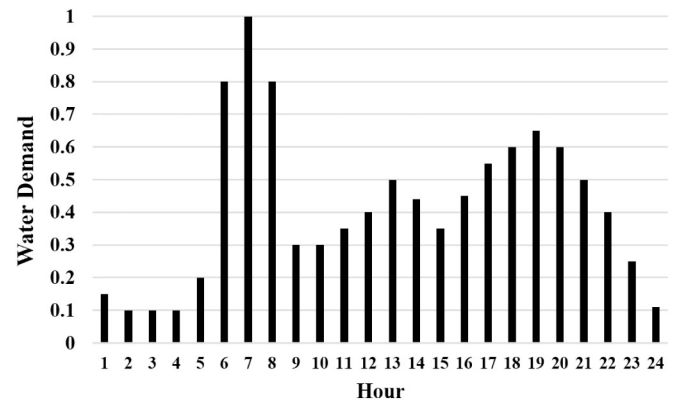


Fig. 7. Hourly normalized water demand.

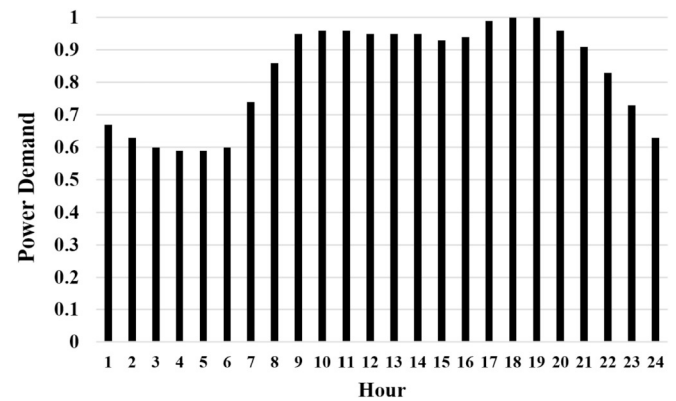


Fig. 8. Hourly normalized electric power demand.

Table 3
Total cost result of the simulations for the both cases.

Cost component (\$)	Base case	Double storage	Difference	
			Value (\$)	%
Start-up	1380	1020	360	26.087
Shut-down	3060	2280	780	25.49
Generation	987,042	940,708	46,333	4.694
Total	993,987	944,008	49,978	5.0281

which is described in the following. Table 4 presents statistics of the loading status of four desalination units in the both cases. The loading means the ratio of the water produced per hour to the unit capacity. According to the 24-hour operation, the loading percent will also have 24 values based on the produced water. The lower this value, the farther it was from its nominal capacity and therefore the less stress will be

Table 2
Data of the water desalination units.

Title		Symbol	Unit	WDU1	WDU2	WDU3	WDU4
Bus location		–	–	4	8	17	20
Technology		–	–	RO	RO	MSF	MED
Feed water		–	–	Seawater	Brackish	Seawater	Seawater
Recovery ratio		–	%	60	70	40	50
Energy Consumption	Electric	$PE_{(d)}^{WDU}$	kWh/m3	4.5	2.2	3.5	2.5
	Thermal	$PT_{(d)}^{WDU}$	MJ/m3	0	0	230	180
Daily production capacity		$WC_{(d)}^{WDU}$	m3/day	400,000	200,000	50,000	80,000
Storage tank pump power	Charging	$WPC_{(d)}^{Cha}$	m3/h	1000	500	200	300
	Discharging	$WPC_{(d)}^{Dis}$	m3/h	1000	500	200	300
Storage tank capacity		$WC_{(d)}^{Tank}$	m3	5000	2500	1000	1500
Daily peak water demand		$Wd_{(d)}^{Peak}$	m3/h	15,000	7000	2000	3000

Table 4
Status of water desalination units loading (%).

Item		WDU1	WDU2	WDU3	WDU4	Average
Max	Base case	90.0	84.0	96.0	90.0	90
	Double storage	84.0	84.0	86.4	81.0	83.85
	Difference	+6.0	0.0	+10.0	+9.0	6.25
Ave.	Base case	37.5	35.0	40.0	37.5	37.5
	Double storage	38.6	36.0	41.2	38.6	38.6
	Difference	-1.1	-1.0	-1.2	-1.1	-1.1
Min	Base case	9.0	8.4	9.6	9.0	9
	Double storage	9.9	9.2	10.6	9.9	9.9
	Difference	-0.9	-0.8	-10.0	-0.9	-3.15

on it. Three first part rows of the table show the highest loading value each unit experienced during the day for the both cases. As the results show, the maximum loading is reduced considerably at the expense of a negligible increase in the minimum and average loading. Between the

desalination units, the WDU2 has the least impact due to the unit's low energy consumption. On the contrary, the WDU3 has the highest impact with 10% reduction in the maximum loading. According to the results, each unit is influenced proportional to its energy consumption level. The last column of the table demonstrates that average maximum loading of all units is reduced by 6.25% at the expense of 1.1 and 3.15% growth in average and minimum loading of the units, respectively.

Fig. 9 depicts hourly water generated by each WDU in four parts. Each part demonstrates generated water for the both cases. As in the figures, except for the WDU2, generated water at main peak of the day is reduced. The water needed at that hours is stored previously at the tank by increasing water production at low demand periods, i.e., hours 1–5. Moreover, it can be seen from the figures that the second peak of the water demand, which is smaller and occurs between hours 16–21, is also reduced for all units.

Charging and discharging waters to the storage tanks for all desalination units are demonstrated in Table 5. In the table, positive and

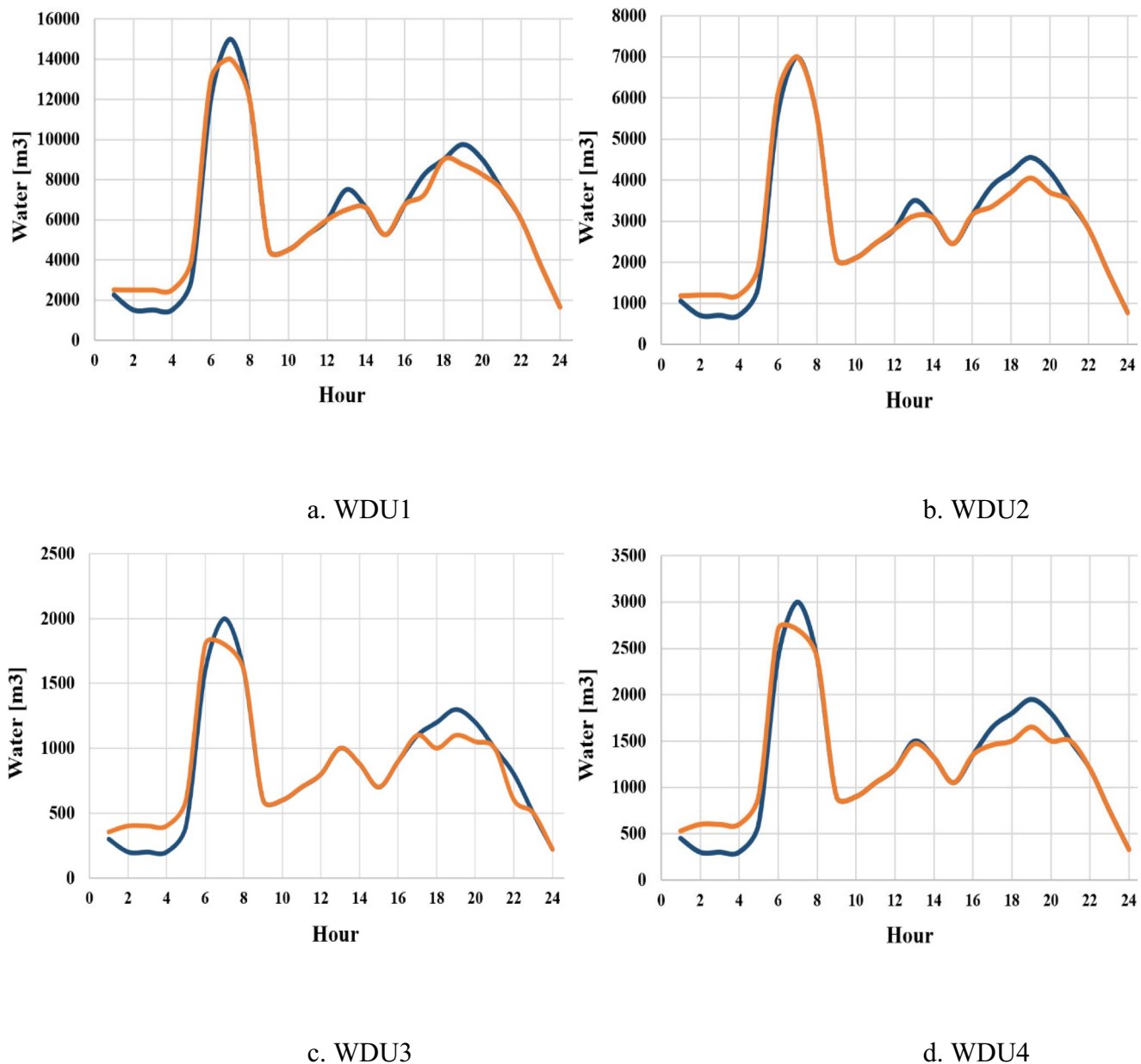


Fig. 9. Hourly generated water (blue = base case and orange = double storage). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Hourly charging and discharging waters of the WDU storage tanks (m³).

Hour	WDU1	WDU2	WDU3	WDU4
1	263	131	52	79
2	1000	500	200	300
3	1000	500	200	300
4	1000	500	200	300
5	1000	500	200	300
6	1000	500	200	300
7	-1000	-500	-200	-300
8	0	0	0	0
9	0	0	0	0
10	0	0	0	0
11	0	0	0	0
12	0	0	0	0
13	0	-500	0	-112.295
14	0	0	0	0
15	0	0	0	0
16	0	0	0	0
17	-750	-500	-200	-300
18	-1000	0	-200	-300
19	-1000	-375	-93.757414	-300
20	-1000	-500	-200	-112.705
21	0	0	0	0
22	0	0	-56.242586	0
23	0	0	0	0
24	0	0	0	0

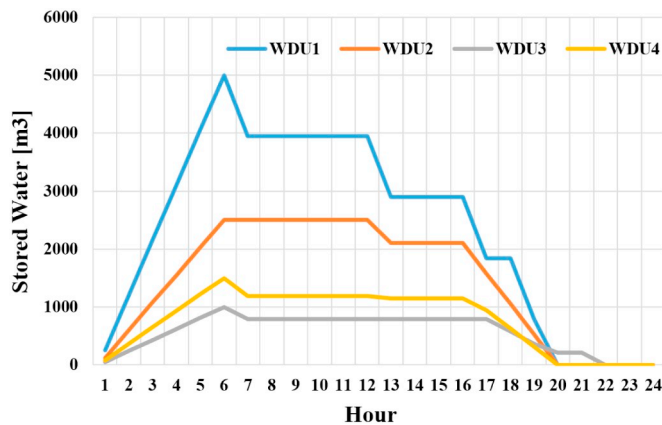


Fig. 10. Water stored in the tank for each WDU.

negative values stand for charging and discharging waters, respectively. The following are the highlights of the results.

The first one is that all of the WDU have been worked with the full capacity of the power of the charging pump. In addition, they have all tried to charge in the early hours of the day. This is because there is the minimum demand for water during these hours. Discharging profile also demonstrates a similar behavior for all of the WDUs. As in the figure, the charged water is discharged at three different time periods. The first one is the main water peak demand of the system at the early morning. The second is the third rank of the water demand at 13 o'clock. Accordingly, the last one is the second water

Table 6
Statistics of the on-line generating units for the both cases.

Item		Base case	Double storage	Diff.
Online units	Min	13	13	0
	Average	20.54	18.41	2.13
	Max	23	20	3
	Total	493	442	51
Start-ups		10	7	3
Shut-downs		14	10	4

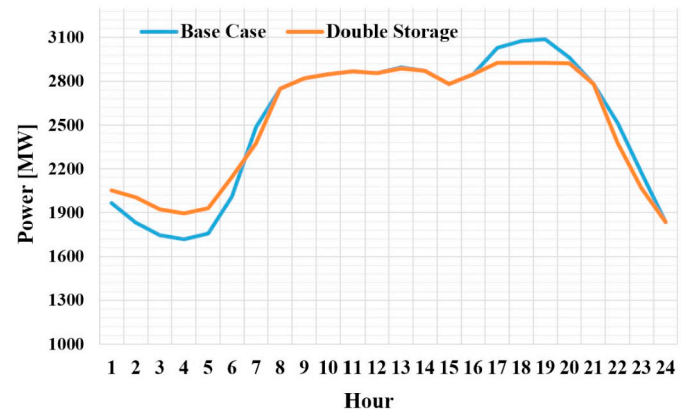


Fig. 11. Total hourly produce power by the generating units.

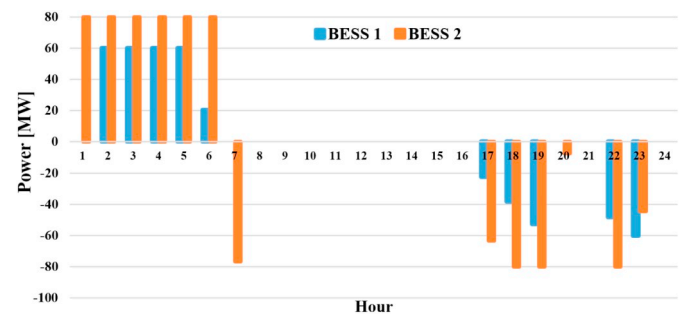


Fig. 12. Hourly charge/discharge status of the batteries.

peak of the system during hours 17–20. Also, there was a little discharge for WDU3 at hour 22. Similar to the charging profile, all of the WDU have been used all of the discharging pump capacity to drain the tank faster.

Status of the stored water in the storage tank of the WDUs is presented in Fig. 10. As it could be concluded from the charging profile of the units, the water is mainly stored at initial hours of the day. After that, the stored water is gradually drained to supply the demand peaks at various time periods.

Besides the WDUs, the generating units are also benefited from the joint operation of the storage units, namely battery and water tank. Table 6 reports statistics of the on-line units during operation for the both cases. The table states that minimum number of the on-line units is not changed by using both of the storage systems. However, other indices are changed including total number of on-line units and their average and maximum. As in the table, in the peak period operation of the system, total number of on-line units is decreased by 3. In addition, in the whole daily operation period, the total number of on-line units has been reduced by 51 by using storage systems together. This is equal to a net 2.13% reduction in the hourly number of on-line units. This reduction in the number of on-line units is results to reduction in start-ups and shut-down by 3 and 4 time, respectively. This will be result to less wear and tear of the generating units, in turn. Fig. 11 demonstrates the total power produced by all of the on-line units at any time period for the both cases. As it can be concluded from the figure, peak of the electric load profile is shaved effectively by utilizing storage systems. This peak shaving is achieved by the valley filling at the periods with low electric energy demand.

At last but not the least, Fig. 12 shows the charging/discharging profile of the both batteries. As one can easily have concluded from the figure, both of the batteries have been drawn charging power from the connected bus at the early hours of the day, i.e., during hours 1–6. This is because of the lowest level of the electric power demand and opportunity to full the battery pack with the lower energy cost. The

energy stored with the cheap price is injected to the network at the peak power periods, namely hours 17–23. Values of the charging and discharging powers demonstrate that the system has been used the complete capacity of the power electronic interface of the batteries almost at all of the working periods. This means that if more power rating of the batteries were available, the program could achieve better results in terms of cost savings.

4. Conclusions

In this paper, a new model for the simultaneous scheduling of water desalination units and thermal power plants is introduced. The proposed model considered internal water storage tank of each desalination unit as well as the grid storage batteries. In addition, the model takes into account constraints on the transmission network, i.e. transmission lines and buses. The results of the proposed model implementation on a sample system showed a significant reduction in the

total cost of operation owing to the simultaneous scheduling. Strictly speaking, total daily operation cost has fallen by more than 5%. In addition, the maximum loading rate of desalination units decreased significantly (by an average of 6.25%). In addition, the number of start-ups and also shut-downs of the thermal units has been reduced, which in turn has reduced the associated costs as well as increasing the life-time of the units. The model proposed in this paper can be used for future studies, for example, modeling and evaluating effect of network uncertainties, optimal investment planning of the joint power-water supply networks, renewable energy integration in the desalination units, and reliability and resiliency assessment of the power-water supply.

Declaration of competing interest

There is not any conflict of interest.

Appendix A. Appendix

A.1. Ramp modeling of the water production

Similar to the power plants, water generated by the desalination units may be limited by the generation ramp. In this case, the difference in water produced between two consecutive hours cannot be more or less than a certain amount. These limitations can be mathematically modeled as follows. In Eq. (a1), $Wg_{(d)}^{RU}$ stands for ramp-up limit of each water desalination unit. This inequality states that increase in the water generation from the previous time period cannot exceed ramp-up limit of the WDU. Similarly, in Eq. (a2), $Wg_{(d)}^{RD}$ denotes ramp-down limit of each water desalination unit. This inequality also implies that decrease in the water generation from the previous time period cannot be more than the ramp-down limit of the corresponding WDU.

$$Wg_{(d,h)}^{WDU} - Wg_{(d,h-1)}^{WDU} \leq Wg_{(d)}^{RU} \quad \forall d \in \Omega_D, \quad h \in \Omega_H \quad (a1)$$

$$Wg_{(d,h-1)}^{WDU} - Wg_{(d,h)}^{WDU} \leq Wg_{(d)}^{RD} \quad \forall d \in \Omega_D, \quad h \in \Omega_H \quad (a2)$$

A.2. Data of the generating units

Data for the generating units connected to the system in Fig. 6 can be found in the following table [31].

Table A2

Data for the generating units [31].

#	Bus	Cost coefficients			Capacity (MW)		Down time (h)	Up time (h)	Initial state	Ramp (MW/h)
		Quadratic (\$/MW ²)	Linear (\$/MW)	Fix (\$)	Min	Max				
1	1	0	130	400.7	16	20	1	1	0	180
2	1	0	130	400.7	16	20	1	1	0	180
3	1	0.014142	16.1	212.3	15.2	76	4	8	0	120
4	1	0.014142	16.1	212.3	15.2	76	4	8	0	120
5	2	0	130	400.7	16	20	1	1	0	180
6	2	0	130	400.7	16	20	1	1	0	180
7	2	0.014142	16.1	212.3	15.2	76	4	8	0	120
8	2	0.014142	16.1	212.3	15.2	76	4	8	0	120
9	7	0.052672	43.7	781.5	25	100	8	8	1	420
10	7	0.052672	43.7	781.5	25	100	8	8	1	420
11	7	0.052672	43.7	781.5	25	100	8	8	1	420
12	13	0.00717	48.6	832.8	69	197	10	12	0	180
13	13	0.00717	48.6	832.8	69	197	10	12	0	180
14	13	0.00717	48.6	832.8	69	197	10	12	0	180
15	15	0.328412	56.6	86.4	2.4	12	2	4	0	60
16	15	0.328412	56.6	86.4	2.4	12	2	4	0	60
17	15	0.328412	56.6	86.4	2.4	12	2	4	0	60
18	15	0.328412	56.6	86.4	2.4	12	2	4	0	60
19	15	0.328412	56.6	86.4	2.4	12	2	4	0	60
20	15	0.008342	12.4	382.2	54.3	155	8	8	1	180
21	16	0.008342	12.4	382.2	54.3	155	8	8	1	180
22	18	0.000213	4.4	395.4	100	400	1	1	1	1200
23	21	0.000213	4.4	395.4	100	400	1	1	1	1200
24	22	0	0	0.001	10	50	0	0	1	600
25	22	0	0	0.001	10	50	0	0	1	600
26	22	0	0	0.001	10	50	0	0	1	600

(continued on next page)

Table A2 (continued)

#	Bus	Cost coefficients			Capacity (MW)		Down time (h)	Up time (h)	Initial state	Ramp (MW/h)
		Quadratic (\$/MW ²)	Linear (\$/MW)	Fix (\$)	Min	Max				
27	22	0	0	0.001	10	50	0	0	1	6000
28	22	0	0	0.001	10	50	0	0	1	6000
29	22	0	0	0.001	10	50	0	0	1	6000
30	23	0.008342	12.4	382.2	54.3	155	8	8	1	180
31	23	0.008342	12.4	382.2	54.3	155	8	8	1	180
32	23	0.004895	11.8	665.1	140	350	48	24	1	240

References

- [1] F.R. Rijsberman, Water scarcity: fact or fiction? *Agric. Water Manag.* 80 (1–3) (2006) 5–22.
- [2] D. Seckler, R. Barker, U. Amarasinghe, Water scarcity in the twenty-first century, *International Journal of Water Resources Development* 15 (1–2) (1999) 29–42.
- [3] H. Bouwer, Integrated water management: emerging issues and challenges, *Agric. Water Manag.* 45 (3) (2000) 217–228.
- [4] D.P. Loucks, Sustainable water resources management, *Water Int.* 25 (1) (2000) 3–10.
- [5] L.S. Pereira, T. Oweis, A. Zairi, Irrigation management under water scarcity, *Agric. Water Manag.* 57 (3) (2002) 175–206.
- [6] E. Jones, et al., The state of desalination and brine production: a global outlook, *Sci. Total Environ.* 657 (2018) 1343–1356.
- [7] M. Qasim, et al., Reverse osmosis desalination: a state-of-the-art review, *Desalination* 459 (2019) 59–104.
- [8] Amin Ghobeity, Alexander Mitsos, Optimal design and operation of desalination systems: new challenges and recent advances, *Current opinion in chemical engineering* 6 (2014) 61–68.
- [9] Mohammad Ali Abdelkareem, et al., Recent progress in the use of renewable energy sources to power water desalination plants, *Desalination* 435 (2018) 97–113.
- [10] Nikolay Voutchkov, Energy use for membrane seawater desalination—current status and trends, *Desalination* 431 (2018) 2–14.
- [11] H. Shemer, R. Semiat, Sustainable RO desalination—energy demand and environmental impact, *Desalination* 424 (2017) 10–16.
- [12] Mehrjerdi H., et al., Daily-seasonal operation in net-zero energy building powered by hybrid renewable energies and hydrogen storage systems *Energy Conversion and Management* 201, 112156.
- [13] A.J. Wood, B.F. Wollenberg, G.B. Sheblé, *Power Generation, Operation, and Control*, John Wiley & Sons, 2013.
- [14] G.B. Sheble, G.N. Fahd, Unit commitment literature synopsis, *IEEE Trans. Power Syst.* 9 (1) (1994) 128–135.
- [15] N.P. Padhy, Unit commitment—a bibliographical survey, *IEEE Trans. Power Syst.* 19 (2) (2004) 1196–1205.
- [16] S. Sen, D.P. Kothari, Optimal thermal generating unit commitment: a review, *Int. J. Electr. Power Energy Syst.* 20 (7) (1998) 443–451.
- [17] Q.P. Zheng, J. Wang, A.L. Liu, Stochastic optimization for unit commitment—a review, *IEEE Trans. Power Syst.* 30 (4) (2015) 1913–1924.
- [18] M. Farshad, et al., Coordinated scheduling of power generation and water desalination units, *IEEE Trans. Power Syst.* 34 (5) (2019) 3657–3666 Sept. 2019.
- [19] H. Saboori, et al., Energy storage planning in electric power distribution networks—a state-of-the-art review, *Renew. Sust. Energ. Rev.* 79 (2017) 1108–1121.
- [20] H. Mehrjerdi, Simultaneous load leveling and voltage profile improvement in distribution networks by optimal battery storage planning, *Energy* 181 (2019) 916–926 15 August 2019.
- [21] A.D. Khawaji, I.K. Kutubkhanah, J.-M. Wie, Advances in seawater desalination technologies, *Desalination* 221 (1–3) (2008) 47–69.
- [22] F.S. Pinto, R.C. Marques, Desalination projects economic feasibility: a standardization of cost determinants, *Renew. Sust. Energ. Rev.* 78 (2017) 904–915.
- [23] Cipollina, Andrea, Giorgio Micale, and Lucio Rizzuti. "Seawater desalination." *Conventional and Renewable Energy Processes*. Berlin & Heidelberg: Springer-Verlag (2009).
- [24] H. Mehrjerdi, et al., Modeling and optimal scheduling of battery energy storage systems in electric power distribution networks, *J. Clean. Prod.* 234 (2019) 810–821 10 October 2019.
- [25] R. Hemmati, H. Saboori, Short-term bulk energy storage system scheduling for load leveling in unit commitment: modeling, optimization, and sensitivity analysis, *J. Adv. Res.* 7 (3) (2016) 360–372.
- [26] S. Vemuri, L. Lemonidis, Fuel constrained unit commitment, *IEEE Trans. Power Syst.* 7 (1) (1992) 410–415.
- [27] R. Hemmati, H. Saboori, S. Saboori, Assessing wind uncertainty impact on short term operation scheduling of coordinated energy storage systems and thermal units, *Renew. Energy* 95 (2016) 74–84.
- [28] M. Carrión, J.M. Arroyo, A computationally efficient mixed-integer linear formulation for the thermal unit commitment problem, *IEEE Trans. Power Syst.* 21 (3) (2006) 1371–1378.
- [29] Yong Fu, et al., Integration of large-scale offshore wind energy via VSC-HVDC in day-ahead scheduling, *IEEE Transactions on Sustainable Energy* 7.2 (2015) 535–545.
- [30] H. Saadat, *Power System Analysis*, McGraw-Hill Companies, 1999.
- [31] C. Grigg, et al., The IEEE reliability test system-1996. A report prepared by the reliability test system task force of the application of probability methods sub-committee, *IEEE Trans. Power Syst.* 14 (3) (1999) 1010–1020.
- [32] R.E. Rosenthal, *GAMS: A User's Guide*, GAMS Development Corp, 2017.
- [33] I.L.O.G. Cplex, 12.4 User's Manual, ILOG SA, Gentilly, France, 2015.